

1 OS structures

NOTE:

Throughout the exam, "the company" refers to the hypothetical start-up company introduced in the first question.

Throughout the exam, "the designed OS" or "the OS" refers to the hypothetical Unix-like OS introduced in the first question.

If a start-up selects a Unix-like OS for its supercomputer, what kernel architecture does it use?

Select one alternative:

- ☐ Layered kernel
- ☐ Microkernel
- ☐ Hybridkernel
- ☐ Monolithic Kernel

The Unix-like OS is divided into two fundamental modes. What are these two modes?

Select one alternative

- ☐ Application mode and kernel mode
- ☐ Application mode and system mode
- ☐ User mode and system mode
- ☐ User mode and kernel mode

Maximum marks: 3

2 Process and Thread

The company intends to use the abstraction of processes as in Unix-like OSs. Which of the following is **TRUE** about processes?

Select one alternative:

- ☐ A system call
- ☐ A program stored on the disk
- ☐ A program in execution
- ☐ A block of memory

The designed OS has three fundamental states: ready, running, and blocked. If a process issues an I/O request, to which state does it transition?

Select one alternative

- ☐ Running to Ready
- ☐ Ready to Running
- ☐ Blocked to Ready
- ☐ Running to Blocked

Unlike Unix-based systems, Windows uses a system call different from Unix to create a new process. Which system call does Windows use for process creation?

Select one alternative

- ☐ exec()
- ☐ fork()
- ☐ pthread_create()
- ☐ createprocess()

What is the purpose of the wait() system call in Unix-based OSs?

Select one alternative

- ☐ To pause the execution of the parent process until a child process finishes
- ☐ To wait for the creation of child process
- ☐ To replace the current process with a new program
- ☐ To create a new process with a new program

The target device has a multi-core CPU, so the company also implements threads as in Unix-like OSs. What is the fundamental difference between a process and a thread?

Select one alternative

- ☐ A process shares memory with other processes, while a thread does not.
- ☐ A process is a program in execution, while a thread is a subset of a process.
- ☐ A process is managed by the CPU, while a thread is managed by the user.
- ☐ A thread has its own address space, while a process does not.

Which of the following is **NOT** shared among threads of the same process?

Select one alternative

- ☐ Global variables
- ☐ Code
- ☐ Heap
- ☐ Stack

Maximum marks: 9

3 Scheduling

The company's supercomputer operates in two scenarios: 1) a limited number of applications with known execution times; and 2) an unknown number of computation-intensive and interactive applications.

For the first scenario, which **non-preemptive** scheduling algorithm is optimal for minimizing the average **waiting time** ?

Select one alternative:

- ☐ STCF
- ☐ SJF
- ☐ FIFO
- ☐ Round-Robin

For the second scenario, which scheduling algorithm is best suited to optimize average **response time** for interactive applications?

Select one alternative

- ☐ STCF
- ☐ Priority scheduling
- ☐ RR
- ☐ SJF

Multi-level feedback queue (MLFQ) scheduling combines features of which two scheduling algorithms?

Select one alternative

- ☐ Priority scheduling and RR
- ☐ SJF and RR
- ☐ FIFO and RR
- ☐ SJF and FIFO

In MLFQ scheduling, how does the scheduler decide when to move a process to a lower-priority queue?

Select one alternative

- ☐ When the process exceeds its time quantum in the current queue
- ☐ When the process completes its execution
- ☐ When the process uses less CPU time than expected
- ☐ When the process requests I/O

Maximum marks: 6

4 Memory

The designed OS uses virtual memory to manage physical memory on the supercomputer. What is a virtual address?

Select one alternative:

- ☐ An address generated by the CPU that is mapped to a physical address
- ☐ The address of the hard disk storage location
- ☐ The actual physical location in main memory where data is stored
- ☐ The address used exclusively for I/O devices

The company considers using the dynamic relocation method to manage virtual memory. Which of the following is **FALSE** about the dynamic relocation method?

Select one alternative

- ☐ It determines the physical addresses of a process during execution.
- ☐ It allocates variable-size memory blocks to processes as needed.
- ☐ It uses a limit/bound register to store the memory size of a process.
- ☐ It allocates fixed memory blocks to processes.

Managing memory in a continuous manner can lead to memory underutilization, so memory segmentation is used in some OSs to improve memory utilization. Which of the following statements about segmentation is **FALSE**?

Select one alternative

- ☐ Segmentation divides a program's memory into logical units such as code, data, and stack.
- ☐ Each segment has a base address and a limit that defines its size.
- ☐ Segmentation eliminates all external fragmentation.
- ☐ Segmentation allows different segments to grow or shrink independently.

When allocating a memory space to a segment, different memory allocation algorithms can be used. Which memory allocation algorithm assigns the largest available partition for the process?

Select one alternative

- ☐ Best-Fit
- ☐ Next-Fit
- ☐ Worst-Fit
- ☐ First-Fit

Maximum marks: 6

5 Paging

Memory segmentation suffers from external fragmentation and is difficult to manage, so the company decides to use paging to manage virtual memory. Which of the following is **TRUE** about paging?

1. Paging can lead to internal fragmentation
2. Each page in a virtual address space is mapped to multiple frames in physical memory
3. Pages and frames in a system have different sizes
4. In paging, translation information is stored in a page table

Select one alternative:

- ☐ 2,4
- ☐ 1,4
- ☐ 1,3
- ☐ 2,3

The company first evaluates the inverted paging method for their OS. Which of the following is **TRUE** about inverted paging?

1. Inverted paging reduces the size of the page table by maintaining a single entry per physical frame
2. Inverted paging makes page sharing difficult
3. An entry in the inverted page table stores a page number and offset
4. Inverted paging is faster than traditional paging

Select one alternative

- ☐ 2,4
- ☐ 2,3
- ☐ 1,2
- ☐ 1,3

After evaluating inverted paging, they decide to use traditional paging, which uses a separate page table for each process. In this paging method, which of the following statements about the traditional paging are **FALSE**?

Select one alternative

- ☐ The page size can be different for each process.
- ☐ The size of the page table depends on the virtual address space and page size.
- ☐ The page table maps virtual page numbers to physical frame numbers.
- ☐ Paging suffers from internal fragmentation.

NOTE: In our context, the terms "page number" and "page index" refer to the same concept.

The supercomputer has a 64-bit CPU and the default page size is set to 32KB. How many bits are used for the offset?

Select one alternative

- ☐ 12 bits
- ☐ 13 bits
- ☐ 15 bits
- ☐ 14 bits

It is found that a larger page size is more likely to result in wasted memory due to internal fragmentation, so the OS reduces the page size to 4KB. Additionally, it implements a linear page table and uses only 48 bits for virtual addresses. How many bits are used for the page number?

Select one alternative

- ☐ 27
- ☐ 28
- ☐ 36
- ☐ 39

When a process is initialized, its page table is loaded into main memory. As a result, each memory access requires two accesses to main memory. A small cache, called the translation lookaside buffer (TLB), is added to accelerate the translation of virtual to physical addresses. Which of the following is **TRUE** about TLB?

Select one alternative

- ☐ The TLB stores only physical addresses.
- ☐ The TLB stores the entire page table of one process.
- ☐ The TLB stores copies of frequently used page table entries.
- ☐ The TLB stores some frequently accessed pages for the currently running processes.

If the TLB is full, a page replacement algorithm is used to replace some of the entries in the TLB. Which of the following algorithm mimics the optimal replacement algorithm?

Select one alternative

- ☐ Round-Robin
- ☐ RANDOM
- ☐ Least-recently used
- ☐ FIFO

In addition to slow address translation, a linear paging table also results in a larger page table size. It is recommended to implement a multi-level page table. If each page table entry takes 8 bytes, how many entries can

fit in a single page of 4KB? The OS implements a 3-level page table. How many bits are used for each level of the page directory?

Select one alternative

- ☐ 1024, 10 bits
- ☐ 512, 10 bits
- ☐ 1024, 9 bits
- ☐ 512, 9 bits

Maximum marks: 12

6 Concurrency

Which of the following is **NOT** a necessary condition for solving the critical section problem?

Select one alternative:

- ☐ Progress
- ☐ Mutual Exclusion
- ☐ Bounded Waiting
- ☐ Preemption

Which of the following statements about condition variables is **FALSE**?

Select one alternative

- ☐ Condition variables store shared data between threads.
- ☐ Condition variables are always used together with a mutex to avoid race conditions.
- ☐ A thread waiting on a condition variable will automatically reacquire the associated mutex before returning from wait().
- ☐ Condition variables are used to block a thread until a particular condition is met.

If a semaphore is initialized to 2 and its current value is -1, how many threads are waiting for the semaphore?

Select one alternative

- ☐ 3
- ☐ 2
- ☐ 1
- ☐ 0

Maximum marks: 4.5

7 I/O devices

What is the advantage of Direct Memory Access (DMA)?

Select one alternative:

- ☐ Reduces CPU overhead by allowing devices to transfer data directly to memory
- ☐ Requires no device drivers
- ☐ Increases CPU utilization by polling devices constantly
- ☐ Slows down I/O operations for better synchronization

How does I/O interrupt work in an OS?

Select one alternative

- ☐ The CPU constantly checks the status of the I/O device until it finishes.
- ☐ The I/O device signals the CPU when it has completed its task, suspending the current process if needed.
- ☐ The OS ignores I/O operations until all processes are finished.
- ☐ The CPU directly controls I/O devices without any signaling.

Which disk scheduling algorithm reorders requests to minimize disk head movement?

Select one alternative

- ☐ Priority scheduling
- ☐ Round-Robin
- ☐ SSTF (Shortest Seek Time First)
- ☐ FIFO (First-In-First-Out)

Which of the following is an advantage of Solid-State Drives (SSDs) compared to Hard Disk Drives (HDDs)?

Select one alternative

- ☐ Lower cost per gigabyte
- ☐ Greater mechanical durability due to moving parts
- ☐ Higher read/write speed
- ☐ Larger maximum storage capacity

A 15000 RPM hard disk drive (HDD) receives a request to read a block of data from a random location on the disk.

What is the average total latency (in milliseconds) for this operation?

Assume:

- **Average seek time:** 9 ms
- **Rotational speed:** 15000 RPM

- **Transfer time:** negligible (0 ms)

(Note: how to calculate the average rotation latency)

Select one alternative

- ☐ 12
- ☐ 14
- ☐ 13
- ☐ 11

Maximum marks: 7.5

8 Files

Which of the following stores the information of a file system?

Select one alternative:

- ☐ Data block
- ☐ Boot block
- ☐ Inode table
- ☐ Superblock

Inode is a data structure to represent a file in OS, and it includes the metadata of the file. Which of the following is **NOT** stored in an inode?

Select one alternative

- ☐ Filename
- ☐ Timestamps (creation, modification)
- ☐ File ownership
- ☐ File Size

What is a file descriptor in an operating system?

Select one alternative

- ☐ A unique identifier for a file stored on disk
- ☐ A human-readable name for a file
- ☐ A data structure that stores the entire content of a file
- ☐ A small integer used by a process to access an open file or I/O resource

If a process opens a file using the `open()` system call, the return value of `open()` is 3. How many files has been opened by the process **before this call**?

Select one alternative

- ☐ 1
- ☐ 0
- ☐ 3
- ☐ 2

What is the role of the bitmap in a file system?

Select one alternative

- ☐ It maps file names to their physical locations.
- ☐ It keeps track of free and used disk blocks.
- ☐ It stores file metadata such as permissions and timestamps.
- ☐ It manages all files store on the disk.

In a file system similar to the **very simple file system** discussed in the lecture and the textbook, if a process calls `open()` to open a file `"/usr/test.txt"`, in what order are the following components accessed?

1. usr inode block
2. test.txt inode block
3. usr data block
4. test.txt data block

Select one alternative

- ☐ 1 → 3 → 2 → 4
- ☐ 2 → 1 → 4 → 3
- ☐ 1 → 2 → 3 → 4
- ☐ 3 → 1 → 4 → 2

Maximum marks: 9

9 Process

Each blank is worth 2 points.

The OS has a shell application, a command-line interpreter widely used in Unix-like OSs which is used to execute a new command.

Since the OS is Unix-like, it uses the three system calls, `fork()`, `exec()`, and `wait()` to manage processes.

The shell process is considered the parent process, and executing a new command involves creating a child process from the parent.

The shell must wait for that new command to complete before proceeding.

First, fill in the blanks with the three system calls, **`fork()`**, **`exec()`**, and **`wait()`**, in the following code snippet to complete the shell application. Arguments for the `exec()` function can be skipped.

% indicates a comment

Shell Process:

```
int main() {
    int count = 0;
    while (TRUE) {
        if (gets () == NULL) % gets() is a function used to receive input the new command.
            continue;
        if (  == 0) {
            
        } else {
            
            count ++;
        }
    }
}
```

The shell maintains a counter, **`count`**, to track how many new commands it executes. Does this `count++` work as planned?

- A. Yes. Because it is in the parent process, it is incremented every time a new command is executed,
- B. Yes. Because it is in the child process, it is incremented every time a new command is executed,
- C. No. Since it is in the parent process, it will not be executed.
- D. No. Since it is in the child process, it will not be executed.

Maximum marks: 8

10 Scheduling

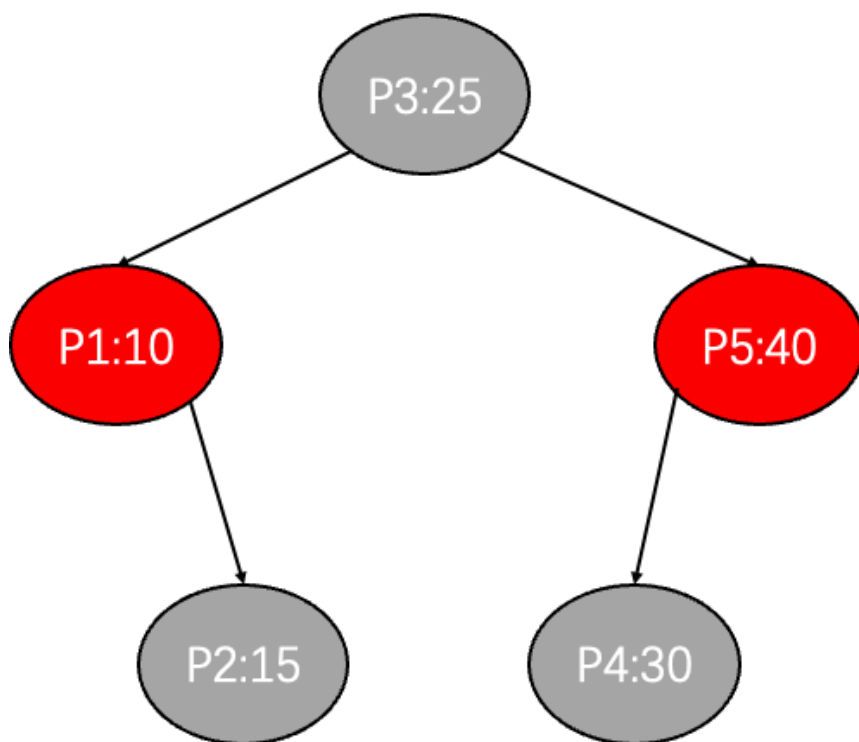
Each blank is worth 2 points.

Completely Fair Scheduling (CFS) is the default scheduling algorithm used in Linux.

The objective of CFS is to guarantee fairness among all running processes. CFS use virtual runtime to determine the schedule of processes.

In addition to virtual runtime, each thread or process in CFS also has a **nice value**. What does this nice value in CFS determine? A thread or process's (one word)

What is the complexity of selecting the next scheduled thread or process in CFS?



CFS uses a red-black tree to implement the scheduling algorithm.

Each node in the data structure stores the virtual runtime of a process.

Px:Y : Px is the process number, while Y denotes the property of Px.

The figure above shows five processes currently running on the system. Which processes will be scheduled last?

CFS allocates time slices to processes, similar to RR. If some processes complete, what happens to the allocated time slices of the remaining processes? The time slices become .

If a process with a virtual time of 12 resumes after an I/O request, what will its virtual value be based on the current status? (Assume that no processes in the structure have accumulated virtual runtime.)

11 Paging

Each blank is worth 2 points.

The OS uses paging to manage virtual memory. The system has a 64-bit CPU and a multi-level paging.

Virtual address: 39 bits.

Page size: 4KB

Page table structure: 3 levels

Each level uses 9 bits of the virtual page number

Split the virtual address **0x7FA3B6C29D584E12** into three level page directory indices and an offset. The least significant bits are used for address translation. **Please answer them in hexadecimal numbers.**

L3	L2	L1	Offset

The following are some page-to-frame mappings.

Page number	physical frame
0x0EA	0x3000
0x10A	0x1345
0x0F5	0x2000
0x18A	0x1000
0xFFF	0x23BF
0x184	0x5ABC
0xADB	0x31CD

What is the physical address of **0x7FA3B6C29D584E12**?

Answer it in hexadecimal number.

Maximum marks: 10

12 Concurrency

Each blank is worth 2 points.

The system implements a ticket dispenser application for a bank.

There is one service desk and 5 seats available for consumers.

If a seat is available, a new consumer can take a ticket and wait for service.

The application involves two types of threads:

Bank staff thread: Processes one consumer's request at a time.

Customer threads: Take tickets and wait to be served at the desk.

Four semaphores are used to implement this application and successfully synchronize information among the threads.

Please fill in the following blanks to complete the code snippet.

1. Initialize three semaphores, empty, full, and mutex.

2. Add three semaphore operations in the bank_staff thread to manipulate three semaphores. Please ensure the correct order of the three operations.

semaphore empty = ; // Tracks available seats (initially all seats are empty)

semaphore mutex = 1; // Ensures mutual exclusion when accessing the ticket dispenser

semaphore full = ; // Tracks the number of occupied seats by waiting customers

semaphore service = 0; // Indicates when a customer is ready to be served. Assume that the bank only has only one staff

Thread customer {

;

sem_wait(mutex);

get_a_ticket();

;

;

;

Be_served();

}

Thread bank_staff {

while{True} {

sem_wait(full);

sem_post(empty)

sem_post(service)

serve_customer();

}

}

Maximum marks: 12

13 File systems

Please briefly describe how to delete an inode in a file system. **(3 points)**

Note: Your answer should address how a file and its inode is processed in deletion.

Fill in your answer here

Maximum marks: 3

14 Comments

You can write some comments for your solutions here; these will not be graded.

Fill in your answer here

Maximum marks: 0